

A Review of Domain Knowledge Representation for Robot Task Planning

Xiaolei Sun

College of Intelligence Science and Technology,
National University of Defense Technology
109 Deya Avenue, Kaifu District,
Changsha, China
00-86-17605106121
sunxiaoleijc@163.com

Yu Zhang

College of Intelligence Science and Technology,
National University of Defense Technology
109 Deya Avenue, Kaifu District,
Changsha, China
00-86-13975856486
redarmy_zy@163.com

ABSTRACT

The review analyzes and defines the related key concepts firstly. Then the article reviews the knowledge representation (KR) formalism and the existing knowledge systems in robotics. The review divides KR in robotics into three categories, namely semantic networks, rules and logical knowledge representation. Then the existing knowledge systems are classified according to the three categories. Finally the article introduces three typical knowledge representation languages in the application of robot task planning and the corresponding task planners. It is still a disadvantage that the domain knowledge of the existing task planners is generated manually, so that the article introduces the method of machine learning to generate the domain knowledge automatically. Our contributions are to facilitate the selection of a knowledge system according to the requirements of robotic application and predict the development trend, hoping to inspire and promote the research in autonomous robotics.

CCS Concepts

• Information systems → Data management systems → Database design and models

Keywords

Knowledge representation; Robot task planning; Symbolism AI; Knowledge engineering; Automatic planning

1. INTRODUCTION

There are three main methods to implement Artificial Intelligence (AI): Symbolism, Connectionism, and Actionism. [1]

Symbolism is an intelligent imitation method based on logical reasoning, also known as Logicism. Its principle is mainly the physical symbol system hypothesis and the principle of limited rationality. It has been dominant in AI for a long time. Symbolism believes that AI comes from mathematical logic. They believe that the basic unit of human cognition and thinking is the symbol, and the cognitive process is an operation on the symbolic representation and reasoning. Symbolism is devoted to imitate the cognitive

process of human with the computer symbolic operation. It inputs such symbols into computers capable of processing symbols, thus it imitates the cognitive process of human beings and realizes AI. [2] The typical representative technology is Knowledge Engineering (KE). The development of the symbolic AI method in KE is introduced later in this chapter.

Connectionism is based on neural network and connection mechanism between networks and learning algorithm. Deep learning is the typical representative technology.

Actionism is an action intelligence imitation method based on "perception-action". It believes that action is a combination of various physical responses that organisms use to adapt to environment. Its theoretical goal is to anticipate and control action. Reinforcement learning (RL) is the typical representative technology.

These theories have driven the development of AI greatly, which have formed their own unique problem-solving system and examples in practice in different periods. Comparatively speaking, each method has its own characteristics and advantages. Symbolism can express and reason knowledge, and it is interpretative; Connectionism has the characteristics of inductive learning; Actionism has the feedback mode of traditional control theory. For the robot task planning, the AI technology dominated by symbols and KE has stronger and more suitable applicability. Next, we will give an introduction to the development process of KE.

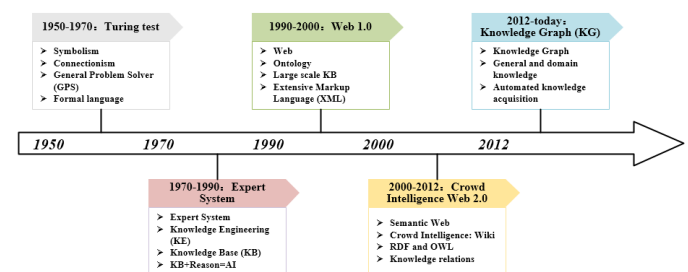


Figure 1 The development process of KE.

The development process of KE is shown in Figure 1. From the 1950 to 1970, the KR methods of this period mainly included logical knowledge representation, the knowledge of rules and semantic networks. A representative work at this period is the General Problem Solver.

From the 1970 to 1990, AI gradually began to build knowledge-based intelligent systems, advocated intelligence through knowledge and inference engines, and emerged many successful

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domain-specific expert systems, such as the DENRAL expert system and XCON expert system. Prof. Feigenbaum, the winner of the Turing Award in 1994, proposed the definition of KE in the 1970s and established the core position of KE in AI. There has been a new evolution in KR during this period, including frameworks and scripts. [3]

During the period from 1990 to 2000, many large-scale knowledge bases were built manually, such as the widely used English knowledge base WordNet, the common knowledge base Cyc of logical knowledge representation, and the Chinese knowledge base Hownet. Web1.0 connects various knowledge with each other so that it can be shared by the public easily, which promotes the knowledge turns from close to open. [4] In this period, KR theory based on ontology was also proposed.

The third period is 2000-2012. The development of the World Wide Web (WWW) makes various source of knowledge linked with each other. Through the relevance of the WWW, knowledge was constantly generated, thus giving birth to swarm intelligence (the technology which can gather the average persons' intelligence), such as the Wikipedia. In 2001, Tim Berners-Lee who is the inventor of WWW put forward the concept of Semantic Web, aimed at structured semantic representation of Internet information. He also put forward the Internet semantic identity language: RDF (Resource Description Framework) and OWL (Ontology Description Language), which are used to describe the ontology of Internet content. This method obtains through semantic identification of web pages, so as to obtain the semantic information of web and enable the collaboration between humans and machines.

Since 2012, the most important achievement is Knowledge Graph (KG). As known to all, Knowledge is power. The goal of this era is to transform web content into machine-understandable and computational knowledge that powers the AI. In recent years, with the emergence of large-scale structured and semi-structured information in the network and the progress of Natural Language Processing (NLP) and Machine Learning (ML), the acquisition of large-scale knowledge has made great progress. The typical representative was the concept of KG proposed by Google in 2012. Since then, KG has been widely used in semantic search, intelligent recommendation and intelligent question answering and so on. The most representative KGs include DBpedia, Freebase, KnowItAll, and YAGO, as well as BabelNet, ConceptNet, DeepDive, NELL, Probase, Wikidata, XLORE, zhishi.me, etc.

However, what is the relationship between knowledge and robots and how can the knowledge be used to robots? Next, we will introduce the knowledge and the application in robot task planning, according to the existing KR knowledge systems in robotics.

2. ANALYSIS AND DEFINITION OF KEY CONCEPTS

What is the knowledge? Since we will use the concepts related to knowledge in robotics in the rest of the article widely, it is necessary to make the concepts explicit. In the domain of philosophy, and computer sciences, there is no a unified definition of knowledge. Allen Newell's famous Knowledge Level [5] can be seen as a milestone. He thinks that knowledge is a bridge between agents and goals, actions and bodies. Symbol level deals with representation and inference, however, the knowledge level deals with language and entailment.

In our research work, we define knowledge as the following: knowledge is composed of a large number of facts and rules, and

all of the information is correlated, which is meaningful for the task execution of robots. What we mean by meaningful is behavior that is likely to be interpreted as purposeful.

In the development of KE, there are many theories and techniques, and a large number of terminology or concepts. In the research of KE, some earlier appeared concept has evolved to be accepted by most researchers. However, there is still no unified and accurate conceptual interpretation of some new terms proposed in recent years. After a review of a large number of literatures, this article expressly explains and defines these concepts and its boundaries in KE.

As shown in Figure 2, Ontology is conceptual model which can provide the specific domain knowledge for computer systems [6]. Corresponding to ontology, semantic network is a technical method which can describe ontology in some formal language. When ontology is instantiated and complex attributes and relationships are expanded (such as temporal and spatial relationships), it constitutes a semantic network. A single semantic network can express the relationships and attributes between concepts or entities in a specific domain, but it has not reached a certain order of magnitude, which is not enough to call it KG. In this article, we define that KG is a large-scale semantic network which is a significant database organizing the human knowledge about the real world in a structured form. The knowledge in KG is represented as concepts, entities, relationships and attributes [7].

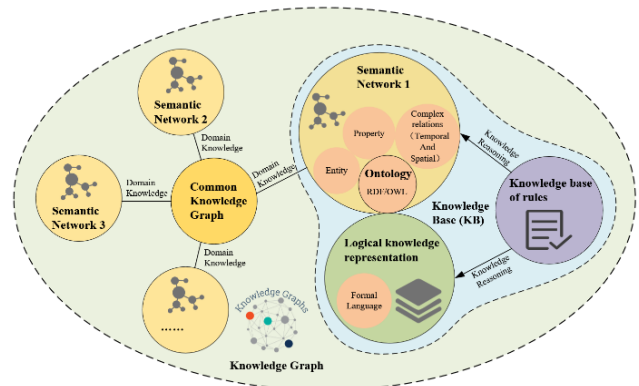


Figure 2 The analysis of concepts related to KE.

KG consists of commonsense knowledge and domain knowledge (see Figure 2). Commonsense knowledge provides the retrieval of commonsense concepts, which corresponds to human cognition of commonsense concepts. Domain knowledge is composed of knowledge in specific domain, such as medical knowledge base, military knowledge base, e-commerce knowledge base, etc. The traditional research on knowledge often goes beyond the preset knowledge boundary. Therefore, the introduction of commonsense knowledge is very important. Knowledge can also be represented through logical language and rules. Problems in the real world can be solved through the rule-based knowledge reasoning.

2017 has been dubbed the "First year of AI" by publications such as the Wall Street journal, Forbes and Fortune. From deep learning and reinforcement learning to neural networks, from Google AlphaGo to Microsoft fifth-generation xiaobing, AI has made great progress. The traditional symbolist AI is difficult to satisfy the expression and application of large-scale knowledge information resources. In view of this, the concept of KG is proposed to express and apply large-scale knowledge information and break

through the bottleneck of the development of traditional symbolist AI.

KG is now widely used in semantic search, intelligent recommendation and question-and-answer systems. In robotics, such KG of large-scale knowledge information are not yet available deeply. Nowadays the traditional KE, such as semantic network and logical language representation, is mostly used in robot task planning. The domain knowledge of robot task planning is still at the stage of manual compilation, and the KG of such large-scale knowledge information defined in this article has not been applied adequately.

3. KNOWLEDGE REPRESENTATION OF AUTONOMOUS ROBOTICS

This chapter summarizes the formalism of knowledge representation in robotics, as well as analyses and summarizes the existing knowledge systems for KR in robotics.

3.1 Knowledge and robotics

In fact, the introduction of knowledge to robots can be regarded as a salient feature of cognitive robots. This session mainly puts more attention on analysis of what is the knowledge of robots and what are the important characteristics of KR.

Firstly, the article puts more attention on high-level robotic control systems which can generate a sequence of actions of high-level tasks in the real world with human beliefs and desires. Our research does not focus on the low-level action drivers.

The purpose of knowledge representation and reasoning is to design a computer system capable of machine-reasonable and machine-interpretable of problems existing in the real world. Knowledge-based systems have computable models in domain of interest, where symbols represent objects or concepts that exist in the real world [8]. The study of knowledge can cover anything in the real world as well as any hypothetical system for computable purposes and knowledge representation. The knowledge-based system manages a knowledge base, which represents and stores the knowledge symbol information of the computable model in a formal language, and reasons by operating these symbols. Through the knowledge base, the system can solve the problem according to the domain-related reasoning [6].

Symbolist AI believes that the intelligent control of a robot can be completed through knowledge representation and knowledge reasoning. For an intelligent robot, it needs to complete the actions assigned by it, and then it needs to cooperate or compete with other agents (such as other robots or humans).

Therefore, there are no more than two kinds of knowledge to be expressed: 1) Knowledge representation related to the execution of a complex high-level task and atomic actions, 2) Knowledge representation related to cooperation and competition with other agents [4]. Knowledge representation related to the execution of a complex high-level task and atomic actions refers to all the objects, concepts and abilities related to task such as the environment map, objects to be recognized and robot capabilities, etc.

3.2 The formalism of knowledge representation in Robotics

The most common formalism of knowledge representation in robotic knowledge systems are based on semantic networks, rules and logic. The semantic networks uses graphical metaphors to visualize conceptual structures. The rules express statements of the real world about a domain with the form as if-then. The logic

uses logical language to express a precise formal semantics for both semantic networks and rules. In this session, we will give an overview of the three formalism of knowledge representation in robotics.

3.2.1 Semantic Networks

Initially, semantic networks derives from the “existential graphs” put forward by Charles Peirce in 1896, which represents logical sentences as graphical node-and-link diagrams [3]. Later, similar concepts were introduced, such as the concept graphs [1], which are syntactically and semantically slightly different.

ConceptNet is a semantic network database consisting of commonsense knowledge which now contains 1.6 million edges and more than 300 000 nodes. Nodes are semi-structured English fragments, which is interrelated by twenty semantic relations. A partial snapshot of ConceptNet in is shown in Figure 3.

Semantic networks are particularly well suitable for expressing taxonomic structure at the conceptual level and expressing relationships and attributes related to the domain of interest. In addition, the representation of a large number of concrete individuals or even data types, such as numbers or strings, does not quite fit the semantic networks. The modern product of semantic network is descriptive logic, which provides strict formal semantics for network structure.

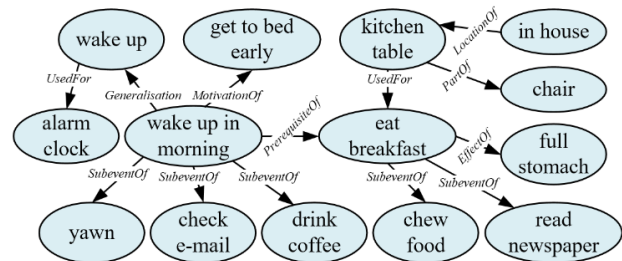


Figure 3 A partial snapshot from ConceptNet’s semantic network, which is in the scenario of household. Take from [9].

3.2.2 Rules

In some domain of interest, rules are another form of knowledge representation that expresses conditions and results. Rules can represent knowledge in the form of if-then constructs, allowing complex derivation statements to be expressed. Rules can be used in logic programming systems, such as language Prolog [10] and deductive databases [4].

The following is an example of a set of rules representing with if-then constructs about the household environment.

1. **IF** two rooms share a door **THEN** the two rooms are connected
2. **IF** someone in a room sharing a door with another room **THEN** someone can enter another room
3. **FACT** Tom is in room 2302 and can enter the room2301

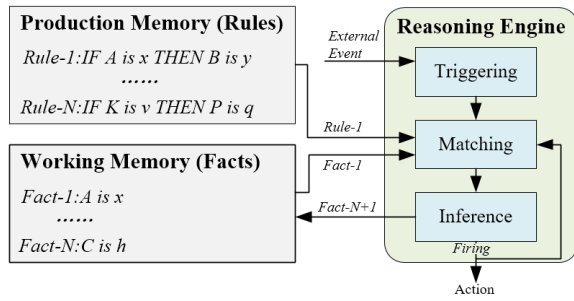


Figure 4 Rule-based System of [11], taken from [11].

Rule is a declarative method to represent knowledge. Rule-based system usually uses ontology model to express the fact knowledge in order to facilitate the reasoning process. The description of knowledge in rule-based system usually includes two aspects, one is the representation of facts in the domain, and the other is the representation of rules based on facts. The system described in [11] is used for worker–cobot agile manufacturing in industrial production. This is defined in as two types of memory, working memory and production memory (see Figure 4). After the external task is input into the system, every time the facts and rules are matched successfully, the internal reasoning mechanism is triggered, and the THEN conclusion will be inferred, which leads to the new fact updating in the fact database.

3.2.3 Logical knowledge representation

Similar to the knowledge representation of semantic networks, many natural language statements can be represented in logical language. For the domain of interest involved in these logical sentences, the KR system need to simply select the appropriate predicates and function notations. Concepts are mapped to unary and relationships are mapped to binary predicates. [2]

Logic is a form of knowledge representation. The most prominent and basic logical formalism in classical KR is "first-order predicate calculus", which is also called first-order logic (FOL) for short. The paper [2] chooses this form for knowledge representation. FOL allows the domain of interest to be described as consisting of objects [1].

In addition to the FOL knowledge representations, there are other logical representations, such as Logic Programs (LP), Description Logics (DL). DL and LP [12] which in turn is similar to the Horn fragment of FOL. Both are then related to def-Horn.

Figure 5 describes the relationships between the various logical knowledge representation methods. DL and Horn are subsets of FOL which is decidable strictly. LP intersects with FOL, but is not fully contained in FOL. For example, FOL can express disjunction that is not expressed in LP. Several of the expression features of LP, which are often used in practical rule-based applications, are expressible in FOL.

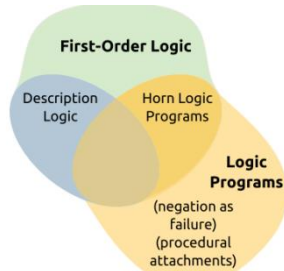


Figure 5 Expressiveness overlap of DL and LP based on Horn clauses, taken from [4].

The goal of the article [4] is to understand the relationship between DL and LP based KR formalisms, so as to combine both. Since DL is similar to a subset of FOL, they focus on the Horn FOL fragment and def-horn which contains no function symbols. This article establishes a new corresponding logical representation called Description Horn Logic (DHL). DHL is included in the intersection, and closely related to the Description Logic Programs (DLP), which can be seen as having a moderate weakening as to the conclusion can be drawn. Both the KR formalism semantic networks and rules can be formalized with logical knowledge representation [2].

3.3 Existing systems for knowledge representation in robotics

This section first clarifies the criteria of the systems for KR in robotics selection, and then introduces representative studies for each category. Table 1 lists the typical KR systems we investigated.

Table 1. The typical KR systems we investigated

Knowledge system	Authors (Institution)	Formalism of KR	Methods	Year	Main reference
KnowRob	Tenorth, Beetz	Section 3.2.1	PROLOG + OWL-DL	2009	[13]
RoboBrain	Saxena (Cornell/Stanford)	Section 3.2.1	Graph + RaQueL	2014	[14, 15]
RoboDB	Juarez A, Hu J,	Section 3.2.1	Semantic Web	2011	[16]
OUR-K	Lim, et al. (Hanyang Univ.)	Section 3.2.3	DL + Horn Clauses	2007	[17]
ORO	Lemaignan, Séverin	Section 3.2.3	DL+RDF	2013	[18]
KeJia	Chen. (USTC)	Section 3.2.3	ASP	2010	[19]
BWIBots	Khandelwal Piyush	Section 3.2.3	BC	2017	[20]
BRMS	Sadik A R, Urban B.	Section 3.2.2	Rule-based +ACL	2017	[11]

We confine ourselves to the knowledge systems as the following described:

1. KR systems based on the real world (including the representation of physical objects in the environment, capabilities and requirements, and tasks)
2. Online and dynamic knowledge acquisition and reasoning
3. Used to solve a certain kind of problems (such as task planning, natural language understanding, and behavior prediction)

We have investigated a large number of knowledge systems and classified them according to the knowledge representation described in section 3.2, as shown in table 1. This chapter will introduce a detailed description of the typical systems in each category. The main contents of investigation include knowledge representation, knowledge reasoning and knowledge application.

3.3.1 The category of Semantic Networks

This classification includes many knowledge systems, such as KnowRob [13], RoboEarth [21], robhow [22], OpenEASE [23],

and so on. The most representative is KnowRob. This session will introduce the KnowRob based on semantic network.

KnowRob is an integrated knowledge management system for autonomous robots. KnowRob aims at the construction of indoor service robot knowledge base, which is composed of ontology, entities and extensible reasoning engine. Researchers can apply the KnowRob in the field of automatic robotics. It is built with a lot of modules, (see Figure 6), which describes the core reasoning system. The core module interfaces with external modules through the Java/Prolog or C/Prolog APIs. KnowRob can update automatically and share knowledge online (see Figure 7), and can be used to solve the corresponding research problems of robots in uncertain environments [24].

KnowRob is based on OWL ontology which is a commonsense ontology, focusing on the household and kitchen domain. KnowRob also stores and reasons internal knowledge through the Semantic Robot Description Language (SRDL) [26], which allows to symbolically represent the capabilities and requirements of the robot and is used for robot task planning.

In the typical extension components of KnowRob, PROBCOG [27] is a method based on Bayesian networks to provide probabilistic reasoning. The article [28] studied the integration with naive physical reasoning, and the article [29] also carried out an experiment on automatic parsing of Web resources in semi-natural language.

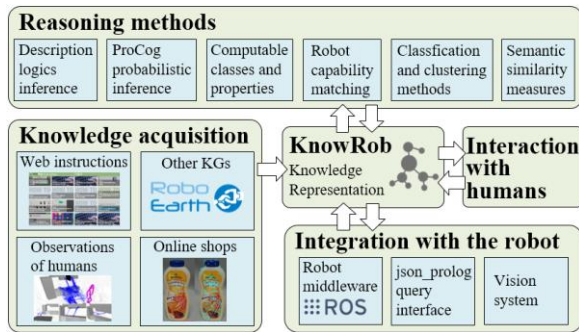


Figure 6 Overview of the KnowRob, taken from [25].

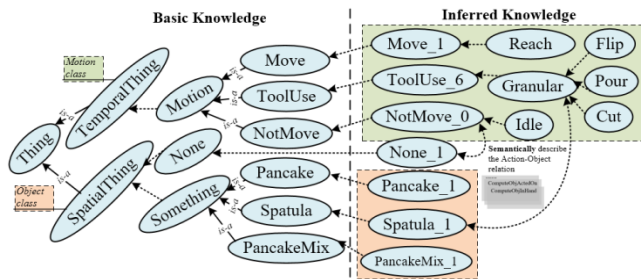


Figure 7 Examples of knowledge updating. On the right, the reasoning engine is used to grow the knowledge base. New nodes are automatically created as needed, taken from [24].

KnowRob provides a mechanism called computables that allows some predicates to be computed by calling external specialized functions.

Combined with Prolog's lazy evaluation strategy, this approach is highly scalable. KnowRob has been deployed on several scenarios, on the robot PR2 and a 2-arm robotic manipulator with the project "assistive kitchen" [30]. These experiments include retrieval and

automatic parsing of recipes, getting and manipulation of various kitchen tools, and collaboration between robots.

3.3.2 The category of Rules

Many researchers try to build a robot knowledge system by using the ontology model and rule-based technology. Sadik A.R. and Urban b. [11] proposed a distributed control solution to solve the problems in industrial production collaboration, combined with ontology-based multi-agent system (MAS) and rule-based Business Rule Management System (BRMS) (see Figure 8). Remarkably, they use ontologies as tools to represent and understand entities and relationships in industrial production. On the basis of ontology, in their work, knowledge reasoning is carried out in combination with rule knowledge base, so as to update knowledge and solve problems.

In addition, the rules-based solution can not only monitor the status of operational resources to assign tasks such as the workers, cobots and machines, but also track the production status of each worker to address situations such as workers' job assignments. Moreover, if both workers are available and have the same productivity, it is quite successful to solve this more complex situation.

In their study work, they selected a specific experimental scenario, which was composed of two workers working with a cobot. In this scenario, the client can generate the production orders in consideration of the main characters and number of products. Taking the production of centrifugal pump in industrial production line as an example, the product is processed and assembled. When a customer sends a production order, the cobot selects from its storage space and places an appropriate number of product-related parts on a worker's workstation, where the workers can process and assemble the products.

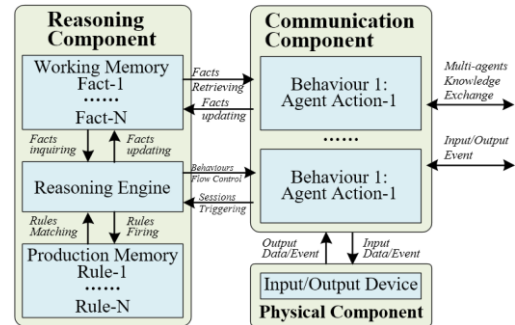


Figure 8 The system based on Rules in [11]

3.3.3 The category of Logical knowledge representation

The KeJia project [19] integrates a knowledge representation language with NLP, and task planning on a mobile platform. KR depends on the Action Language C, which is based on Answer Set Programming (ASP) [1]. These languages are similar to Prolog syntactically which are based on stable logical programming models and support non-monotonic reasoning. For task planning [31] and unspecified NLP, default and non-monotonic reasoning has been specifically studied in KeJia. Amongst other features, the system's NLP capability enables new logical rules to be fetched at run-time.

KeJia has been demonstrated in several scenarios involving human-robot interaction with natural language. These tasks involve many complex actions, such as grabbing, moving, and tipping. The realization of these action tasks is a kind of globally opti-

mized and naive logical reasoning process. Logical reasoning is carried out through the rules given by the system to adapt to more complex scenes. The scenes need to consider changing and mutually exclusive preferences of users.

4. ROBOT TASK PLANNING BASED ON DOMAIN KNOWLEDGE

First of all, we need to explain some concepts. Atomic actions can be executed directly by the robot which have an impact on the environment and its own state. High-level tasks can be decomposed into a sequence of atomic actions. Assuming that the robot has the ability to perform various atomic actions, the complex high-level task can be achieved by constructing a sequence of such atomic actions. For example, a robot may need to hand an object to P1, but may not know the location of P1. However, it may know that it can get the location of P1 by asking P2. Achieving this goal requires a great deal of symbolic knowledge representation and reasoning, as well as interaction with people using GUI, speech recognition, and scene understanding. In addition, in order to achieve these advanced goals, the robot needs to track and update knowledge about the environment, such as the location of P2 and obstacles information. These information is stored in the robot's knowledge base for reasoning and task planning.

Task planning problems for autonomous robotics can be described using many languages according to the review of the recent research works, such as Planning Domain Definition Language (PDDL) [32], Hierarchical Task Network (HTN) [33-35], the action language BC [36] and C+ [37], etc.

In this section, we describe the knowledge representation for robotic task, knowledge reasoning, and task planning. Task planning methods based on robot domain knowledge are summarized according to robot task representation language.

4.1 Task planning with HTN

HTN is an AI method for decomposing high-level tasks. The ultimate planning goal is to decompose a sequence of atomic actions that perform a high-level task. The main idea of HTN planning algorithm is to decompose the input high-level task into some subtasks and atom actions, and then decompose the subtasks until the decomposed to the level of directly executable atomic actions and stop the decomposition. Among them, tasks and subtasks are described by methods in the planning domain, and atomic actions are described by operators [38]. The definition of an atomic action is limited by the precondition and the execution effects, which indicates that under which state the agent can perform the action. The effect of the execution indicates the result of the action on the state table.

Similar to traditional Planning, HTN Planning saves a set of methods and operators in the file of planning domain, as well as the preconditions and results of performing these methods and operations that is, describing the decomposition and execution of tasks. For the file of planning problem of a specific domain, it mainly indicates the initial state when the high-level task instruction is input and the task's goal that need to be achieved.

In the article [39], indoor service robot is taken as the platform and the task of 'water delivery' is taken as an example. HTN method is used to formulate the field description file and problem description file of the task. The methods of the planning domain file are then matched against the execution preconditions of the operation. The matching process is to determine whether the environment information meets the preconditions of a task and subtask

until a series of atomic actions that can complete the task are matched. The goal of task planning is to generate a list of specific atomic action execution that can fulfil the user's request, that is, the matching results, and pass them to the robot to execute.

This project uses JSHOP2 planner to realize robot task planning, and carries out simulation experiment, proving that JSHOP2 planner can generate robot action execution list and realize robot task planning.

4.2 Task planning with PDDL

PDDL is a widely used standard representation language created by McDermott [40] to describe various planning problems. PDDL was developed to simplify the comparison of knowledge systems. PDDL has evolved with multiple iterations of the version over the years, and now it is PDDL Version 3.

An action representing in PDDL is like in a tuple $\langle \text{head}(a), \text{pre}(a), \text{eff}(a) \rangle$ where,

- **head(a)** is the head name of action.
- **pre(a)** is the preconditions of action.
- **eff(a)** is the effect after the execution of action.

Similar to HTN, The input format of PDDL includes two files: domain file and problem file. The domain file represents the operation and the required prerequisites and expected results. The problem file represents the initial state and the desired goal state of the system. The paper [41] studies that both files are automatically generated from the ontology.

4.3 Task planning with BC

In the scenario of household environment, if the two rooms have the same doors then we call the two rooms can be entered into each other directly. We use the R, R_i, and D, D_i to describe the room and the door, respectively.

default $\sim \text{hasAdoor}(R, D)$.

$\text{hasAdoor}(R_1, d_1). \text{hasAdoor}(R_2, d_2). \text{hasAdoor}(R_3, d_3)$.

default $\sim \text{accto}(R_1, D, R_2)$.

$\text{accto}(R_1, D, R_2) \text{ if } \text{hasAdoor}(R_1, D), \text{hasAdoor}(R_2, D)$.

$\text{accto}(R_1, D, R_2) \text{ if } \text{actoc}(R_2, D, R_1)$.

In robot high-level task planning, a robot may need to plan with incomplete information. The article [42] demonstrates how BC can be used for robot task planning to solve the problem. This paper [20] applies BC to the real world of robots, including human-computer interaction to acquire knowledge, optimal task planning to minimize navigation time, and learning for adaptive planning. The paper [20] evaluated a method of using mobile robot Segway to conduct navigation, interact with people and complete tasks such as receiving emails in indoor environment. They also have a test of the method both on simulation environment and physical robot platform.

4.4 Task planning with machine learning

In recent years, researches on robot task planning have given rise to a large number of task planners. However, the domain knowledge of these planners is still programmed manually by human and cannot jump out of the problem of domain-dependence, which is still the bottleneck of robot task planning. It is quite difficult to manually define task planning knowledge. Recent studies [43] have presented methods with ML to generate automatically definition of task planning knowledge.

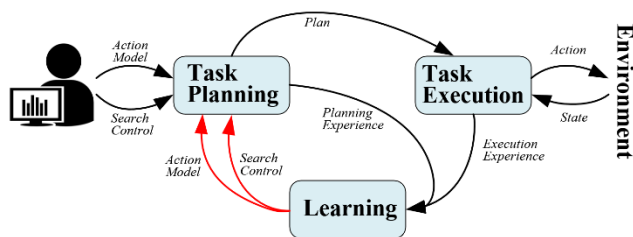


Figure 9 The role of ML in robot task planning.

Even for automatic planning experts, building task planning knowledge models is difficult and time-consuming. Another method is to use ML so that one do not have to manually write a task planning knowledge model. ML in task planning learns in two steps, which are learning the action preconditions and effects [44]. Current domain-independent heuristics algorithms are expensive to compute and planners spend most of time evaluating useless nodes. Because the domain-independent technologies in these domains are all based on actions, the planner's search tree becomes very tricky when the number of objects and action parameters reaches a certain size. It makes it difficult for domain-independent planners to apply them to the practical problems. Therefore, ML plays a role in capturing useful task planning knowledge. The role of ML is illustrated in Figure 9.

ML in task planning is often used to accelerate planners with task planning knowledge [45], specific heuristics [46] and rules [47]. It is also used to improve the quality of the task planning, including success rate [48], learning domain models and program operators [49]. A recent review [43] makes new contributions with acquiring control knowledge.

Due to the diversity of environment, the traditional KR are limited in expressiveness, and they require a lot of training to be better. The paper [50] studied the related RL methods. RL methods are more suitable for learning complex decomposable tasks and integrating ML into robot task planning. Their effectiveness in complex tasks planning for robots has yet to be demonstrated. Another problem of RL methods is that the reward function is difficult to specify and estimate.

5. CONCLUSION

The robotic technology based on knowledge representation has made great progress, which also shows the force of knowledge more and more. However, the current researches are still at the level of traditional knowledge engineering, such as ontology and expert system. In the era of big data, with the generation of a large number of knowledge information, the technology for the management and application of such knowledge information becomes more and more important. The birth of knowledge graph can solve the problem well. It is composed of large-scale semantic network and can manage and apply the knowledge information well. In the field of robotics, although some specific robot ontology models have been established, there is no such knowledge system at the level of knowledge graph.

Based on the current researches in robotics, we predict its future research trends:

1. The automatic generation of domain knowledge in robot task planning.
2. The representation and application of knowledge in multi-robots cooperation and competition.
3. The establishment of robot knowledge graph and the fusion of existing knowledge systems.

4. The application of cloud-based KG technology, which will reduce the dependence of specific hardware and is conducive to the research of multi-agents and swarm intelligence.

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